

EDUCATION RESEARCH

Teaching in an Era of Generative Artificial Intelligence

LLM-Generated multiple choice practice quizzes for preclinical medical students

 Troy Camarata,^{1*}  Lise McCoy,^{2*} Robert Rosenberg,³ Kelsey R. Temprine Grellinger,⁴
Kylie Brettschnieder,² and  Jonathan Berman²

¹Baptist University College of Osteopathic Medicine, Memphis, Tennessee, United States; ²New York Institute of Technology College of Osteopathic Medicine, Jonesboro, Arkansas, United States; ³Drexel University College of Medicine, Philadelphia, Pennsylvania, United States; and ⁴Western Michigan University Homer Stryker M.D. School of Medicine, Kalamazoo, Michigan, United States

Abstract

Multiple choice questions (MCQs) are frequently used in medical education for assessment. Automated generation of MCQs in board-exam format could potentially save significant effort for faculty and generate a wider set of practice materials for student use. The goal of this study was to explore the feasibility of using ChatGPT by OpenAI to generate United States Medical Licensing Exam (USMLE)/Comprehensive Osteopathic Medical Licensing Examination (COMLEX-USA)-style practice quiz items as study aids. Researchers gave second-year medical students studying renal physiology access to a set of practice quizzes with ChatGPT-generated questions. The exam items generated were evaluated by independent experts for quality and adherence to the National Board of Medical Examiners (NBME)/National Board of Osteopathic Medical Examiners (NBOME) guidelines. Forty-nine percent of questions contained item writing flaws, and 22% contained factual or conceptual errors. However, 59/65 (91%) were categorized as a reasonable starting point for revision. These results demonstrate the feasibility of large language model (LLM)-generated practice questions in medical education but only when supervised by a subject matter expert with training in exam item writing.

NEW & NOTEWORTHY Practice board exam questions generated by large language models can be made suitable for preclinical medical students by subject-matter experts.

artificial intelligence; large language model; medical education; multiple choice questions; practice questions

INTRODUCTION

Rationale for Studying Practice Questions: Self-Directed Quizzing and Practice Items

To assess student learning, medical schools administer multiple choice question (MCQ) exams. Each test question is called an “exam item,” and the process of writing them is called “item writing.” Medical schools invest extensive resources into training faculty to develop high-quality MCQ assessments of learning. These are high-stakes exams for medical students and are formative for board examinations. Medical students also use MCQs for practice during preparation for exams (1–3). For this reason, providing students with ample practice through formative assessments has become urgent for medical schools. Medical students continually request practice quizzes to solidify learning before summative exams.

Not only do practice test items serve as exam preparation, but medical educators have determined that repetitive quizzing is a powerful learning tool (4, 5). Other medical education

researchers suggest that formative practice prevents failure on exams (6). Preclinical students value formative assessments for the immediacy of feedback (7). Therefore, it is a benefit to complement student learning content, such as lectures, with formative practice items.

Medical school faculty are under increasing pressure to generate MCQ items for frequent formative practice quizzing and peer instruction (8), as well as for course exams. Writing new MCQ assessment items consumes considerable effort, which can lead some faculty to struggle with this task. It takes extensive training of faculty to develop valid and reliable assessments (9). The cost per written item can be in excess of \$100 per item (10). Past efforts at automatic item generation have used methods such as template-based generation, which results in question “families” with similar formats by changing variables in a question, and rule-based generation (11, 12). Studies of faculty-generated items for in-house examinations tend to show poor results for faculty untrained in item writing but items of high quality for trained faculty (13, 14).



*T. Camarata and L. McCoy contributed equally to this work.

Correspondence: J. Berman (Jonathan.Berman@nyit.edu).

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Because board exams are often taken at the end of the second year of preclinical education, each assessment must match the guidelines for exam items from the United States Medical Licensing Exam (USMLE) or Comprehensive Osteopathic Medical Licensing Examination (COMLEX-USA), must be factually accurate, and must reach an appropriate difficulty level and discrimination index, a measure of the ability of the question to discriminate between high-performing and low-performing students. These guidelines are developed by the National Board of Medical Examiners (NBME) and National Board of Osteopathic Medical Examiners (NBOME), respectively.

Several commercial companies provide practice test questions for medical education. These commercial test item banks allow medical students to practice board-style exam items to prepare themselves for USMLE and COMLEX-USA board exams. While these item banks are extensive, many items are too advanced or do not precisely match specific lectures for preclinical course topics. When faculty are not sufficiently trained, using or adapting these items for course exams can result in low-quality items (9). Due to the significant effort required to write original faculty-derived questions, there is a lack of suitable practice items for first- and second-year medical students that are matched to lesson content, precisely the correct level, and/or topic specific. Practice questions developed by (or supervised by) faculty teaching the material may be seen as more useful by medical students and as reflecting questions they are likely to see on exams.

Rationale for Use of Large Language Models

In the last several years there has been a significant increase in the quality of available large language models (LLMs) that are capable of generating text in response to human prompts. Perhaps the best known of these is ChatGPT (OpenAI), which represents a variety of models including GPT-3.5 and GPT-4. These models are capable of producing human readable, relevant text in response to prompts and can generate USMLE-style practice questions when prompted.

Encouraged by earlier experimental work using LLM tools to answer questions on medical board exams (15, 16), the research team chose to explore the abilities of LLMs to assist in item writing. Additionally, interest in the use of LLMs as a tool for item writing has been expressed among medical educators (17). These models have been demonstrated as feasible tools for generating reading passages for other exam types (18).

The promise of LLMs is coupled with concerns about its use. LLMs sometimes produce information that is false or misleading, but formatted as factual, in what is called “hallucination” (19). LLMs also present ethical concerns about inappropriate attribution of LLM-generated content to human authors (20) and potential biases within the LLM output (21).

Item Quality

Poor quality exam items can lead to inaccurate assessment (22) and can even present misinformation to students. Poor quality items used for formative assessment or practice might not adequately prepare students for summative assessments or may not lead to effective learning (23). Therefore, it is

important to understand the quality of exam items produced by LLMs.

The current study’s primary aim is to explore the feasibility of using LLMs such as ChatGPT by OpenAI to assist in generating USMLE/COMLEX-USA-style practice quiz items for medical student use as study aids by evaluating the questions produced for “item writing flaws,” which are a common measure of adherence to board exam question adherence to guidelines.

We hypothesized that the exam items generated by the LLM would contain few enough item writing flaws (as evaluated by 3 independent subject matter experts) that they could be revised to a state where they could be provided to students using “minimal” effort.

MATERIALS AND METHODS

In the fall of their second year of medical school, New York Institute of Technology College of Osteopathic Medicine (NYITCOM) students participate in a course entitled “Principles and Practice of Osteopathic Medicine II (PPOM2),” which includes the majority of their foundational physiology education. The renal physiology module is distinct from the pathophysiology module and comprises 12 video-recorded lectures made available to students each week through a lecture management system. The 12 lectures are structured to approximately follow the presentation order of the 6th edition of Mosby’s Physiology Series, *Renal Physiology* (24), with a final review session to cover “high yield” topics likely to appear on level-1 board exams. The physiology topics are preceded by a lecture to review renal anatomy initially taught in the first year.

The Process of Generating Test Items Using ChatGPT

The researcher (J.B.) generated 10 practice quizzes using GPT-4. Each quiz was based on a specific lecture and aligned to specific learning objectives. Each quiz included 3–10 test items in a standard USMLE clinical vignette format based on the lecture’s learning objectives or clinical conditions mentioned in the lecture. Prompts were designed to specify certain desirable aspects of USMLE-style exam items as defined by the NBME Item Writing Guide. An example prompt is as follows:

“Please help me write a USMLE-style exam item. It should have a clinical vignette, a patient history, lab values (with normal ranges), and a separate question stem. It should not be a pseudocase. There should be five possible answers, one correct and four incorrect but of similar length to the answer. Include a rationale. The topic should pertain to Gitelman Syndrome.”

Certain features were emphasized, such as having five possible answers and including a rationale because without this text in the prompt the questions generated would often only have three distractors or lack a rationale.

This prompt format was developed through a process of trial and error. Because generative AI is trained on datasets including some derived from the internet (19), the practice questions it generates often contain errors that are consistent with those frequently found in human-written cases, such as

pseudocases. As such, the prompt was designed with instructions to avoid some of the most common mistakes that human authors (and thus AI) are apt to make. These prompts were given to GPT-4 (the versions available between April and July of 2023) (25).

If responses were judged to be obviously inappropriate, such as being a clear pseudocase, or not pertaining to the desired topic, either the system was prompted to try again, or the researcher (J.B.) requested a revision based on the nature of the error such as “This is a pseudocase. Please revise.” The iterative process was never repeated more than once per question. For five items, there was an error noticeable on *prima facie* examination. Examples include a clear pseudocase, not addressing the desired topic or an incorrect number of distractors. No question was manually edited.

Pseudocases are items that include a clinical vignette but in which the clinical vignette does not contribute information needed to answer the question (26). An example of a pseudocase item generated by GPT-4 is as follows:

Clinical Vignette:

A 35-yr-old woman with no significant past medical history presents to her primary care physician for an annual check-up. She mentions that she has been feeling well and has no complaints. She exercises regularly, eats a balanced diet, and has no family history of chronic diseases. Physical examination is unremarkable.

Question:

Which of the following is the primary pacemaker of the heart?

- A) Atrioventricular node
- B) Sinoatrial node
- C) Bundle of His
- D) Right atrium
- E) Left ventricle

Assessment of Exam Item Quality

After the class ended, three qualified physiologists with knowledge of the renal system and trained in exam item writing from outside institutions (T.C., R.R., and K.R.T.G.) were selected to act as judges of generative AI-produced exam items. The assessment process was developed independently of the faculty teaching the course and developing the software/exam items (J.B.).

Cognitive level.

Although Bloom’s taxonomy is widely used in education, it has not been empirically validated (27). However, there is no empirically validated and widely accepted framework to assess the cognitive level of MCQs. In lieu of such a framework, Buckwalter’s modification of Bloom’s taxonomy was used (28). Rather than six cognitive “domains,” Buckwalter’s modification lists three cognitive “levels,” where level 1 questions primarily test recall, level 2 primarily tests understanding and interpretation of data, and level 3 tests application of knowledge. Although these are framed as levels, it should not be assumed that a “level 1” MCQ is cognitively easier than a “level 2” MCQ because of the aforementioned lack of empirical validation.

Item writing flaws.

A rubric was developed to assess the adherence of the LLM-generated exam items to best practices, as defined by the

NBME item writing handbook (26). These are factors commonly assessed in studying item writing (29) and are aligned with the taxonomy of MCQ item writing guidelines presented by Haladyna et al. (27). The rubric assessed a variety of factors identified as common item writing flaws, including grammatical errors, factual errors, incorrect number of answer options, negative phrasing, teaching in the stem or vignette, word repetition, overly detailed answer options, convergence errors, and inappropriate distractors. The initial determination of item writing flaws was made by the identification of items flagged by two out of three reviewers. Each item was discussed, and a consensus was reached.

Appropriateness.

The appropriateness of an exam item refers to its psychometric validity as a measure of a test-taker’s ability. Likewise, an NBME-style exam item may follow all item writing rules but not measure knowledge of the desired domain for a variety of reasons, such as misalignment with topic or biases. For an exam item to be appropriate, it should be both valid, measuring what it claims to measure, and reliably able to provide results across time and multiple populations. In this case, the appropriateness of questions for preclinical medical students was assessed with a modified Likert scale. The scale was based on a combination of the item construction (clinical scenario or single-line rote style), alignment with NBME and/or NBOME blueprints, and whether the item was assessing the targeted learning objective with an expected Bloom’s category utilized by three independent medical schools (Doctor of Medicine and Doctor of Osteopathic Medicine). A score of 1 on the scale meant the item was poorly constructed and did not assess a concept suggested by licensure exam blueprints or at the cognitive level expected of medical students, while a score of 5 suggested the item was appropriately constructed, assessed an important clinical concept for medical students, and fit an expected cognitive Bloom’s category for preclinical undergraduate medical education. An intermediate score suggested an item may have few item writing flaws and aligns with board exam blueprints while fitting into Bloom’s categories such as “remember” or “understand.” Final ratings were determined by averaging Likert scores from each of the three expert item reviewers.

Revision classification.

To determine if the exam items generated by LLM were suitable for revision, the subject-matter expert faculty classified each question based on the degree of revision needed to generate a new suitable question. The following rubric was used to determine the classification of revision needed per item (Table 1). If there was misleading/misinformation and multiple content-related issues or writing flaws in the scenario, lead-in, and answer options, the item was classified as very difficult to revise. If there was an item writing flaw in the question scenario and answer options, the item was classified as difficult to revise. If the lead-in, an answer option, and/or minor changes in the scenario were needed, this was classified as “easy.” If only the lead-in, such as revising a pseudocase, or one answer option needed to be changed, this was classified as a very easy revision. Finally, a “no revision needed” class was also scored (Table 1). Three expert item reviewers classified items independently using the

Table 1. Rubric for scoring revision difficulty of exam items

Description	Revision Difficulty
Misleading/misinformation, multiple content problems or writing flaws in scenario and answer options.	Very difficult
Item writing flaw in question scenario and answer options.	Difficult
Lead-in, an answer option and/or minor revision in scenario needed.	Easy
Lead-in, or one answer option requires revision.	Very easy No revision needed

rubric (Table 1). Discrepancies in classification were discussed, and a consensus was reached.

Statistics.

Spearman correlation statistics were calculated between the number of identified item writing flaws for each item or the appropriate difficulty Likert scale value with the revision classification. Revision classification was converted to a numerical score as the following: very difficult or difficult class = 1, easy or very easy class = 2, and no revision needed = 3. Statistical outputs were reported using American Psychological Association guidelines.

RESULTS

MCQ Item Quality Analysis

The 65 MCQ test items were analyzed by subject matter experts trained in item writing to identify item writing flaws, Bloom's taxonomic order, and subjective appropriateness for preclinical medical students.

Table 2 provides a summary of the item quality analysis. Using modified Bloom's taxonomic order, 49% were first order, 51% were second order, and 0% were third order. Of the 65 items, 63 (93%) included a clinical vignette. All test items featured the correct number of answer options (one correct answer and four distractors). None contained grammatical errors. Most (60%) of the items (39) had at least one

inappropriate distractor. Fourteen items (22%) contained factual or conceptual errors. Five items (8%) contained overly detailed answer options. Five items (8%) contained convergence errors. Four items (6%) had word repetition. One item (2%) was phrased negatively. Three items (5%) included an error regarding teaching in the vignette. The mean subjective appropriateness for preclinical medical students was 2.85 out of 5. Overall, 32 items had at least 1 item writing error (49%), 8% had at least 2 item writing errors, and 3% had at least 3 errors (Table 2; Fig. 1).

Three of the 65 questions were removed from the quiz database midsemester because students pointed out errors in the question content.

Revision Classification

Of 65 questions, 42 required no changes or were rated easy or very easy to revise, and 23 questions were deemed either difficult or very difficult to revise (Fig. 2). In total, 59 were deemed suitable for revision. Many required fixes were minor, such as modifying the lead-in, removing a provided diagnosis, or revising some answer options. More difficult to revise were changing questions so that pseudocases were real cases needed to answer the question, unrealistic diagnostic data, or misleading answer options. To support the revision classifications by expert item reviewers, Spearman correlations were conducted between revision class and appropriate difficulty scale [$r(63) = 0.59, P < 0.001$] and revision class and number of identified item writing flaws [$r(63) = -0.49, P < 0.001$]. There was a median positive correlation between revision classification and appropriateness scale while there was a median negative correlation with number of item writing flaws per item (Table 2).

DISCUSSION

Generalizability

These practice items were given in a single course at a single medical school but were analyzed by faculty at multiple

Table 2. Item quality analysis

Measure	Description	
1) Bloom's level	First order: recall Second order: understanding and interpretation Third order: application	49% 51% 0%
2) Includes a clinical vignette		93%
3) Correct number of answer options	1 key and 4 distractors	100%
4) Level appropriateness	For an exam item to be appropriate, it should be both valid, measuring what it claims to measure, and reliably able to provide results across time and multiple populations. Rated on a Likert scale from 1 (highly inappropriate) to 5 (highly appropriate).	2.85
Item writing flaw, % (n)		
5) Grammar errors		0%
6) Negatively structured	Example: "Which of the following is NOT true."	2% (1)
7) Teaching in the vignette	Extraneous information added to the vignette that explains a concept.	5% (3)
8) Word repetition	Same significant term in the question stem appears in the key, not the distractors.	6% (4)
9) Overly detailed answer options	MCQ answer options are too long.	8% (5)
10) Convergence errors	MCQ answer options are too similar.	8% (5)
11) Factual or conceptual errors		22% (14)
12) Inappropriate distractors		60% (39)

MCQ, multiple choice question.

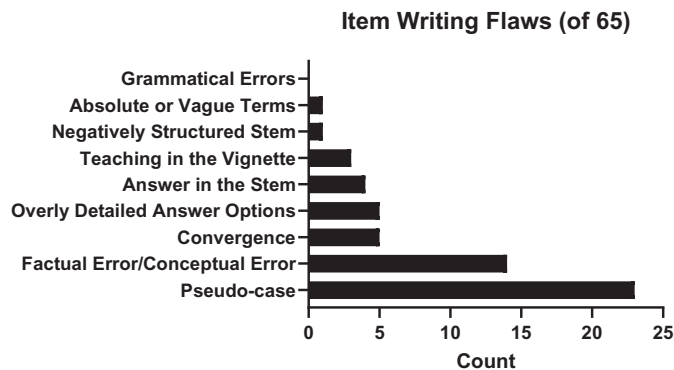


Figure 1. The count of various item writing flaws among the large language model (LLM)-generated practice questions.

medical schools. The testing methods in use during preclinical education will vary between schools. The degree of training in USLME/COMLEX-USA-style item writing will vary between institutions.

Additionally, policies vary between universities for the use of LLM-generated content. Currently, the use of LLM-generated content is disallowed by NBME and NBOME for the generation of questions to be used on actual board examinations. Use of LLM-generated content without disclosure would constitute plagiarism. Thus, while not all institutions may choose to ban the use of LLM-generated content, they may choose to enact specific policies governing its use.

Limitations of LLM-Generated Questions

The primary goal of developing a system for using AI-generated practice questions is to save effort in developing questions by faculty. However, there are several limitations to LLM-generated questions such that the time investment in them is not limited to the time spent entering a prompt. First, since LLMs are known to “hallucinate” (generate false information), questions must be fact-checked by human subject-matter experts. Subject-matter experts must also validate the difficulty level of each question so that they are consistent. Finally, someone trained in exam item writing must review each question to make sure that it aligns with exam item guidelines. Each of these steps requires some time investment. Additionally, because LLMs may not be

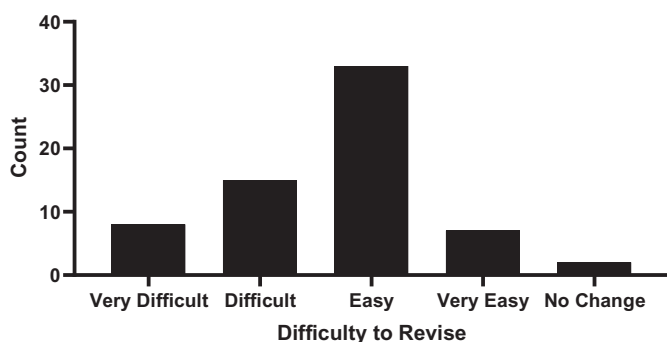


Figure 2. Difficulty of revision classification. Questions were categorized by subject matter experts by their perception of the difficulty of revising each question to meet United States Medical Licensing Exam (USMLE) standards.

immediately updated with the most relevant information, there is a chance that information in the peer-reviewed literature will not be included in practice questions, making questions out-of-date.

Question Quality

The 2.85/5 appropriateness rating of these LLM-generated items and high number of item writing flaws should be considered with reference to the literature on faculty-generated items. Other researchers report that one-fifth of items written by faculty without receiving faculty development for item writing have flaws (29). The subjective quality of items written by faculty untrained in item writing is 2.0/5 (14) but is higher for faculty who underwent faculty development for item writing.

Item Revision

Of 65 exam items, subject-matter experts rated 59 (91%) as “usable templates” and 45 (70%) as needing no revision or being very easy or easy to revise. Given that these exam items are easier to generate than human templates, revision of these exam items may still reduce the time required by faculty to generate practice exam items. Whether items are written by untrained faculty or by an LLM, review by a faculty member with item writing training should improve item quality by eliminating or revising bad items. This approach is similar to introducing “peer review” of items, which has been shown to improve faculty-written items (30).

Conclusions

Exam items generated by LLMs had a large number of item writing and factual errors, requiring human oversight and rewriting for responsible use. LLM-generated questions may be better used as a starting point for human-written questions than as a final product suitable for educational use.

The majority of the questions were suitable templates or rated easy or very easy to revise. Although the process of exam item generation still requires subject matter experts to identify item writing flaws, the items generated have the potential to serve as useful templates, which may save effort compared to writing completely new items from scratch.

DATA AVAILABILITY

Data will be made available upon reasonable request.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

T.C., L.M., R.R., and J.B. conceived and designed research; T.C., R.R., K.R.T.G., and J.B. performed experiments; T.C., L.M., R.R., K.R.T.G., and J.B. analyzed data; T.C., R.R., K.R.T.G., and J.B. interpreted results of experiments; J.B. prepared figures; T.C., L.M., R.R., K.R.T.G., K.B., and J.B. edited and revised manuscript; T.C., L.M., R.R., K.R.T.G., K.B., and J.B. approved final version of manuscript.

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